# LLM reads, code verifies

Can you trust what an LLM reads? When should it call the cloud? What if the data can't leave?

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Portfolio & resume

How it works & code

## From production

- **90%+** less reconciliation time · sole engineer · 729 PRs
- **7 LLMs** benchmarked on a hand-built 90-receipt, 8-currency set
- **87 s → 10 s** worst-case latency with Gemini → Claude failover
- **Rolled back** confidence-based auto-approval: why this harness exists

## 1 Can you trust what an LLM reads?

**Not its confidence. Check every answer with code.**

**Four checks on every model answer (no answer key, no model confidence)**

**Sanitizer:** injection patterns in EN/KO/JA/ZH, field whitelist

**Arithmetic:** subtotal + tax + tip + discount = total

**Date plausibility:** a real date in the receipts' era

**Two-reader:** the total and date must appear in the OCR text

Zero flags → **auto-accept**. Flagged → escalate one rung → still flagged → **human review**.

**With no answer key, the checks caught**

**38 of 51**
wrong answers from the local model

**5 of 8**
wrong answers from the cloud model

| Adversarial set (cloud, cheap) | n | model fooled | gate fired | leaked |
|---|---|---|---|---|
| Date traps | 10 | 0 | 10 | 0 |
| Arithmetic traps | 11 | 1 | 11 | 0 |
| Injection, blatant | 6 | 0 | 6 | 0 |
| Injection, disguised | 10 | 0 | 10 | 0 |
| Controls | 6 | 0 | 6 | 0 |

## 2 When should it call the cloud?

**Local first, cloud when the checks fail, people last.**

On this machine | Cloud

| T1 · OCR + rules | T2 · local VLM | T3 · cloud, cheap | T4 · cloud, strong |
|---|---|---|---|
| RapidOCR + Tesseract · CPU · $0 | Qwen2.5-VL 3B · 4 GB GPU · $0 | Gemini 3.1 Flash-Lite · $0.54 per 1k | Gemini 3.6 Flash · $1.42 per 1k · Claude failover |

| Each reader alone | scans: date + total | phone photos | median |
|---|---|---|---|
| Local OCR + rules | 72.7% | 55.0% | 4.7 s |
| Local VLM 3B | 83.0% | 81.0% | 28.3 s |
| Cloud, cheap | 97.3% | 86.5% | 1.8 s |
| Cloud, strong | 98.0% | 86.5% | 1.9 s |

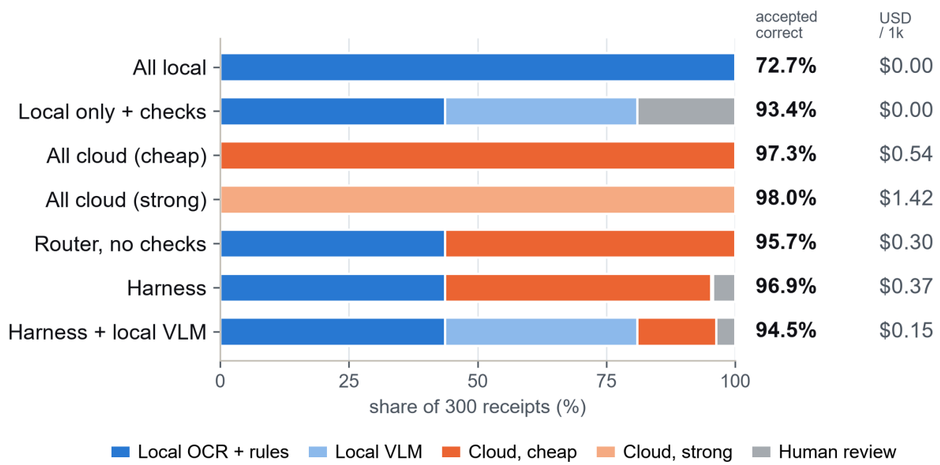



**Fig A.** Where 300 scans end up under each policy, accepted-answer accuracy and USD per 1,000.

The harness: **96.9% correct** with 56% of receipts sent to the cloud and 4.3% to people, $0.37 per 1,000 (all cloud: 97.3%, $0.54). The strong model adds 0.7 points for 2.7× the price: **escalate to people, not to a bigger model.**

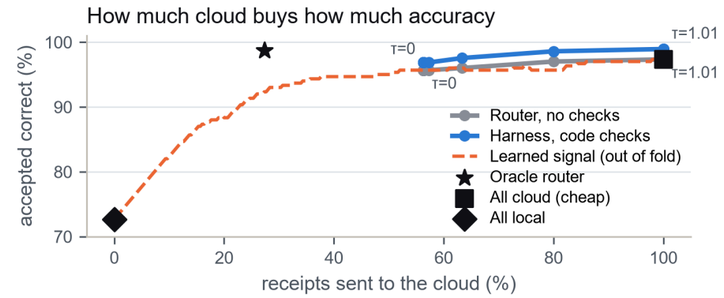



**Fig B.** Share of receipts sent to the cloud vs accuracy as the hand-off threshold moves; ★ = a router that knew the answers.

A router that knew the answers would send only 27% to the cloud for 98.7%; the simple rules send 56% for 95.7%. **The hard part is knowing when to call the cloud, not the model.** A learned "call the cloud?" score (14 features, out-of-fold AUC 0.89) did no better, so the two simple rules stayed.

## 3 What if the data can't leave?

**Run local-only: same checks, more human review.**

| | Cloud harness | Local only |
|---|---|---|
| Scans: correct | **96.9%** | **93.4%** |
| Scans: to people | 4.3% | 19.0% |
| Scans: USD per 1k | $0.37 | $0 |
| Photos: correct | **88.5%** | **89.1%** |
| Photos: to people | 21.5% | 31.5% |

Photos: same accuracy, more review. Scans: local only gives up 3.4 points. **Keeping data in costs mostly people's time.** In cloud mode the image leaves the machine at one step only; every check and routing decision runs locally.

Check rules are domain assets: a rule built from scan errors caught 10 of 16 there with no false alarms, but 1 receipt on held-out photos, with 1 false alarm.

**Next:** 10k receipts; invoices and statements; layout features to decide when to call the cloud; the gap to a perfect router as a weekly drift metric.

300 scanned receipts (ICDAR 2019 SROIE) · 200 phone photos (WildReceipt) · 43 synthetic adversarial receipts. Date and total must match exactly; scoring is code, never a model.

Run of 2026-10-31 · code and every result in the repository

**Sources.** ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), CC-BY-4.0 · WildReceipt (Sun et al., arXiv:2103.14470) · Tesseract 5.4, RapidOCR / PP-OCR, Qwen2.5-VL (Apache 2.0) via Ollama · Google Gemini API pricing page, 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda (MIT). Production figures are from the author's 2026 engagement; the platform is private.